Digital Image Sharing by Diverse Image Media

Kai-Hui Lee and Pei-Ling Chiu

***Abstract*— Conventional visual secret sharing (VSS) schemes hide secret images in shares that are either printed on transparencies or are encoded and stored in a digital form. The shares can appear as noise-like pixels or as meaningful images; but it will arouse suspicion and increase interception risk during transmission of the shares. Hence, VSS schemes suffer from a transmission risk problem for the secret itself and for the participants who are involved in the VSS scheme. To address this problem, we proposed a natural-image-based VSS scheme (NVSS scheme) that shares secret images via various carrier media to protect the secret and the participants during the transmission phase. The proposed (n, n) - NVSS scheme can share one digital secret image over $n - 1$ arbitrary selected natural images (called natural shares) and one noise-like share. The natural shares can be photos or hand-painted pictures in digital form or in printed form. The noise-like share is generated based on these natural shares and the secret image. The unaltered natural shares are diverse and innocuous, thus greatly reducing the transmission risk problem. We also propose possible ways to hide the noise-like share to reduce the transmission risk problem for the share. Experimental results indicate that the proposed approach is an excellent solution for solving the transmission risk problem for the VSS schemes.**

***Index Terms*— Visual secret sharing scheme, extended visual cryptography scheme, natural images, transmission risk.**

I. Introduction

Visval cryptography (VC) is a technique that encrypts a secret image into n shares, with each participant holding one or more shares. Anyone who holds fewer than n shares cannot reveal any information about the secret image. Stacking the n shares reveals the secret image and it can be recognized directly by the human visual system [1]. Secret images can be of various types: images, handwritten documents, photographs, and others. Sharing and delivering secret images is also known as a visual secret sharing (VSS) scheme. The original motivation of VC is to securely share secret images in non-computer-aided environments; however, devices with computational powers are ubiquitous (e.g., smart phones). Thus, sharing visual secret images in computer-aided environments has become an important issue today.

Manuscript received December 11, 2012; revised July 15, 2013 and November 17, 2013; accepted November 17, 2013. Date of publication November 25, 2013; date of current version December 17, 2013. This work was supported by the National Science Council of Taiwan under Contracts NSC101-2221-E-130-018 and NSC 102-2410-H-130-035. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. C.-C. Jay Kuo.

K.-H. Lee is with the Department of Computer Science and Information Engineering, Ming Chuan University, Taoyuan 33348, Taiwan (e-mail: khlee@mail.mcu.edu.tw).

P.-L. Chiu is with the Department of Risk Management and Insurance, Ming Chuan University, Taipei 111, Taiwan (e-mail: plchiu@mail.mcu.edu.tw).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TIFS.2013.2292509

Conventional shares, which consist of many random and meaningless pixels, satisfy the security requirement for protecting secret contents [1]–[4], but they suffer from two drawbacks: first, there is a high transmission risk because holding noise-like shares will cause attackers' suspicion and the shares may be intercepted. Thus, the risk to both the participants and the shares increases, in turn increasing the probability of transmission failure. Second, the meaningless shares are not user friendly. As the number of shares increases, it becomes more difficult to manage the shares, which never provide any information for identifying the shares.

Previous research into the Extended Visual Cryptography Scheme (EVCS) or the user-friendly VSS scheme provided some effective solutions to cope with the management issue [5]–[13]. The shares contain many noise-like pixels or display low-quality images. Such shares are easy to detect by the naked eye, and participants who transmit the share can easily lead to suspicion by others. By adopting steganography techniques, secret images can be concealed in cover images that are halftone gray images and true-color images [14]–[16] However, the stego-images still can be detected by steganalysis methods [17]. Therefore the existing VSS schemes still must be investigated for reducing the transmission risk problem for carriers and shares. A method for reducing the transmission risk is an important issue in VSS schemes.

In this study, we propose a VSS scheme, called the natural-image–based VSS scheme (NVSS scheme), to reduce the intercepted risk during the transmission phase. Conventional VSS schemes use a unity carrier (e.g., either transparencies or digital images) for sharing images, which limits the practicality of VSS schemes. In the proposed scheme, we explore the possibility of using diverse media for sharing digital images. The carrier media in the scheme contains digital images, printed images, hand-painted pictures, and so on. Applying a diversity of media for sharing the secret image increases the degree of difficulty of intercepting the shares. The proposed NVSS scheme can share a digital secret image over $n - 1$ arbitrary natural images (hereafter called natural shares) and one share. Instead of altering the contents of the natural images, the proposed approach extracts features from each natural share. These unaltered natural shares are totally innocuous, thus greatly reducing the interception probability of these shares. The generated share that is noise-like can be concealed by using data hiding techniques to increase the security level during the transmission phase.

The NVSS scheme uses diverse media as a carrier; hence it has many possible scenarios for sharing secret images. For example, assume a dealer selects $n-1$ media as natural shares for sharing a secret image. To reduce the transmission risk, the dealer can choose an image that is not easily suspected as the content of the media (e.g., landscape, portrait photographs, hand-painted pictures, and flysheets). The digital shares can be stored in a participant's digital devices (e.g., digital cameras or smart phones) to reduce the risk of being suspected. The printed media (e.g., flysheets or hand-painted pictures) can be sent via postal or direct mail marketing services. In such a way, the transmission channels are also diverse, further reducing the transmission risk.

In this paper, we develop efficient encryption/decryption algorithms for the (n, n) -NVSS scheme. The proposed algorithms are applicable to digital and printed media. The possible ways to hide the generated share are also discussed. The proposed NVSS scheme not only has a high level of user friendliness and manageability, but also reduces transmission risk and enhances the security of participants and shares.

The remainder of this paper is organized as follows. Section II reviews the research framework. In Section III, we present the proposed NVSS scheme. The encryption/decryption algorithms are proposed in Section IV. In Section V, the security and performance of the proposed NVSS scheme are evaluated by experiments. Finally, we present the conclusions of this work in Section VI.

II. Related Work

Fig. 1 shows the classification of VSS schemes from the carriers' viewpoints. Existing research focuses only on using transparencies or digital media as carriers for a VSS scheme. The transparency shares have either a noise-like or a meaningful appearance. The conventional noise-like shares are not friendly [1]–[4]; hence, researchers tried to enhance the friendliness of VSS schemes for participants [5]–[7]. Generally, simple and meaningful cover images are added to noise-like shares for identification, making traditional VC schemes more friendly and manageable. However, the EVCSs reduce the display quality of the recovered images.

Research has focused on gray-level and color secret images to develop a user-friendly VSS scheme that adds cover images into the meaningless shares [8]–[13]. To share digital images, VSS schemes use digital media as carriers, which makes the appearance of the shares more variable and more user-friendly [13]. Several papers investigated meaningful halftone shares [8]–[11] and emphasized the quality of the shares more than the quality of the recovered images. These studies had serious side effects in terms of pixel expansion and poor display quality for the recovered images, although the display quality of the shares was enhanced. Hence, researchers make a tradeoff between the quality of the shares, the quality of the recovered images, and the pixel expansion of the images.

In another research branch, researchers used steganography techniques to hide secret images in cover images [14]–[16]. Steganography is the technique of hiding information and making the communication invisible. In this way, no one who is not involved in the transmission of the information suspects the existence of the information. Therefore, the hidden information and its carrier can be protected. Steganography has been used to hide digital shares in VSS schemes. The shares in VSS schemes are embedded in cover images to create stego-images. Although the shares are concealed totally and the stego-images have a high level of user friendliness, the shared information and the stego-images remain intercepted risks during the transmission phase [17].

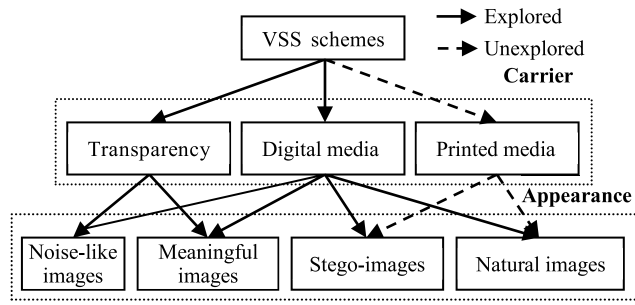


Fig. 1. The classification of the existing VSS research from the viewpoints of carriers.

Recently, Chiu et al. tried to share a secret image via natural images [18]. This was a first attempt to share images via natural images; however, this work may suffer a problem—the textures of the natural images could be disclosed on the share. Moreover, printed images cannot be used for sharing images in the previous scheme.

So far, sharing visual secret image via unaltered printed media remains an open problem. In this study, we make an extension of the previous work in [18] to promote its practicability and explore the possibility for adopting the unaltered printed media as shares.

III. The Proposed Scheme

A. Background

In cryptography, the one-time pad (OTP), which was proven to be impossible to break if used correctly, was developed by Gilbert Vernam in 1917. Each bit or character from the plaintext is encrypted by a modular addition (or a logical XOR operation) with a bit or character from a secret random key of the same length as the plaintext resulting in a ciphertext. The ciphertext was sent to a receiver; then, the original plaintext can be decrypted in the receiver side by applying the same operation and the same secret key as the sender used for encrypting the ciphertext.

As pointed out by Naor and Shamir, the visual secret sharing scheme is similar to the OTP encryption system [1]. In a $(2, 2)$-VSS scheme, the secret random key and the ciphertext that can be treated as two shares in the scheme were distributed to two participants who involve in the scheme. The two participants can decrypt the secret by applying the decryption operation to the shares that were held by the participants.

In this study, we adopt the notion of the OTP technique to share digital visual secrets. Instead of generating a secret random key, we extract the secret key from an arbitrarily picked natural image in the $(2, 2)$-NVSS scheme. The natural image and the generated share (i.e., ciphertext) were distributed to two participants. In decryption process, the secret key will be extracted again from the natural image and then the secret key as well as the generated share can recover the original secret image. The $(2, 2)$-NVSS scheme can be extended to the (n, n)-NVSS scheme by adopting $n-1$ natural images for generating $n-1$ secret keys. In such a way, the visual secret image can be shared by the $n-1$ natural images as well as the generated share.

B. Assumptions

The proposed (n, n)-NVSS scheme adopts arbitrary $n-1$ natural shares and one generated share as media to share one digital true color secret image that has 24-bit/pixel color depth. The objective of this study is to reduce the transmission risk of shares by using diverse and innocuous media. We make the following assumptions:

1. When the number of delivered shares increases, the transmission risk also increases.
2. The transmission risk of shares with a meaningful cover image is less than that of noise-like shares.
3. The transmission risk decreases as the quality of the meaningful shares increases.
4. The natural images without artificially altered or modified contents have the lowest transmission risk, lower than that of noise-like and meaningful shares.
5. The display quality of distortion-free true-color images is superior to that of halftone images.

In the NVSS scheme, the natural shares can be gray or color photographs of scenery, family activities, or even flysheets, bookmarks, hand-painted pictures, web images, or photographs. The natural shares can be in digital or printed form. The encryption process only extracts features from the natural shares; it does not alter the natural shares. The innocuous natural shares can be delivered by participants who are involved in the NVSS scheme, by the owners of the photographs, or via public Internet. Because the natural shares are not altered, it is likely that they will not arouse suspicion during transmission. Even if the natural shares are intercepted, it will not be possible to verify that there is any hidden information in the images before reaching the decryption threshold. In such a scenario, the transmission of the innocuous natural shares is more secure than the transmission of shares in another form, such as noise-like or meaningful shares. Another share, which is generated by the secret image and features that are extracted from $n-1$ natural shares, can be hidden behind other media and then delivered by a well-disciplined person or via a high-security transmission channel.

When the number of shares n increases, based on Assumption 1, the transmission risk of the conventional VSS schemes increases rapidly. On the contrary, regardless of the increasing number of shares, the proposed NVSS scheme always requires only one generated share. Because the natural images have very high security, even though the amount of innocuous natural shares is also proportional to n, the transmission risk of the proposed scheme will increase very slightly as n increases.

In the existing VSS schemes, the types of shares include noise-like shares, shares with binary cover images, and shares with halftone cover images; the latter has the best display quality among the above-mentioned types of shares. Furthermore, the display quality of the proposed true-color natural shares is superior to that of shares with halftone cover images. Based on Assumptions 2 and 3, the transmission risk of the true-color natural shares is the lowest among the existing approaches. Based on Assumptions 4 and 5, the proposed (n, n)-NVSS scheme delivers $n-1$ unaltered natural shares that have a very low transmission risk, this property greatly reduces the transmission cost of delivering $n-1$ natural shares of the scheme. Compared with traditional (n, n)-VSS schemes, which must carefully deliver n noise-like shares, the proposed (n, n)-NVSS scheme must deliver only one generated share in a high-security manner. When the transmission cost is limited, the proposed scheme using unaltered natural shares can greatly reduce transmission risk.

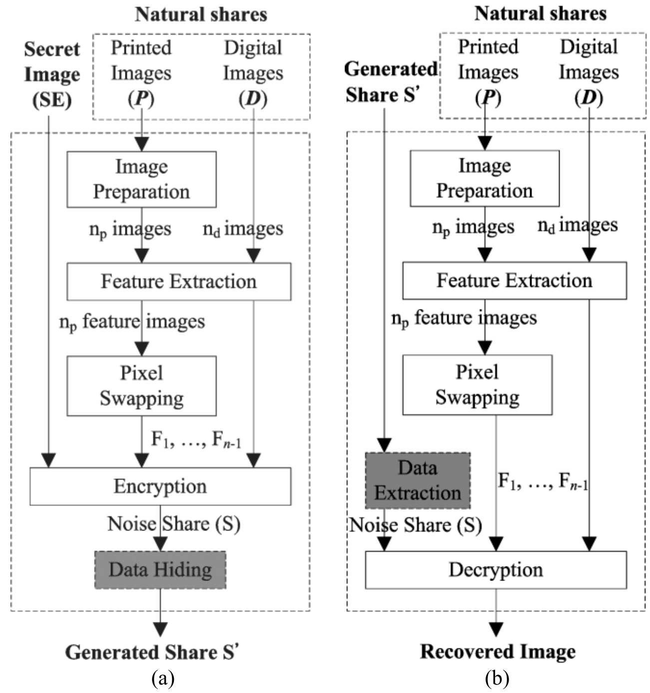


Fig. 2. The encryption/decryption process of the (n, n)-NVSS scheme: (a) encryption process, (b) decryption process.

C. The Proposed (n, n)-NVSS Scheme

As Fig. 2(a) shows, the encryption process of the proposed (n, n)-NVSS scheme, $n \geq 2$, includes two main phases: feature extraction and encryption. In the feature extraction phase, 24 binary feature images are extracted from each natural share. The natural shares $(\mathrm{N}_1, \ldots, \mathrm{N}_{n-1})$ include $\mathrm{n_p}$ printed images (denoted as $\boldsymbol{P}$) and $\mathrm{n_d}$ digital images (denoted as $\boldsymbol{D}$), $\mathrm{n_p} \geq 0$, $\mathrm{n_d} \geq 0$, $\mathrm{n_p} + \mathrm{n_d} \geq 1$ and $n = \mathrm{n_p} + \mathrm{n_d} + 1$. The feature images $(\mathrm{F}_1, \ldots, \mathrm{F}_{n-1})$ that were extracted from the same natural image subsequently are combined to make one feature image with 24-bit/pixel color depth.

In the encryption phase, the $n-1$ feature images $(\mathrm{F}_1, \ldots, \mathrm{F}_{n-1})$ with 24-bit/pixel color depth and the secret image execute the XOR operation to generate one noise-like share S with 24-bit/pixel color depth. Then, to reduce the transmission risk of share S, the share is concealed behind cover media or disguised with another appearance by the data hiding process. The resultant share S' is called the generated share. The $n-1$ innocuous natural shares and the generated share are n shares in the (n, n)-NVSS scheme. When all n shares are received, the decryption end extracts $n-1$ feature images from all natural shares and then executes the XOR operation with share S' to obtain the recovered image, as shown in Fig. 2(b). Each module in Fig. 2 is described in the following sections.

IV. The Proposed Algorithms

A. Feature Extraction Process

This section first describes the feature extraction module that extracts feature images from the natural shares. The module which is the core module of the feature extraction process is applicable to printed and digital images simultaneously. Then, the image preparation and the pixel-swapping modules are introduced for processing printed images.

1) The Feature Extraction Module

There are some existing methods that are used to extract features from images, such as the wavelet transform. However, the appearance of the extracted feature may remain some texture of the original image. It will result in decreasing the randomness of the generated share and eventually reduces security of the scheme. To ensure security of the proposed scheme, we develop a feature extraction method to yield noise-like feature images from natural images such that the generated share is also a noise-like image.

Assume that the size of the natural shares and the secret image are $\mathrm{w} \times \mathrm{h}$ pixels and that each natural share is divided into a number of $\mathrm{b} \times \mathrm{b}$ pixel blocks before feature extraction starts. We define the notations as follows:

- b represents the block size, $\mathrm{b} \in$ even.
- N denotes a natural share.
- (x, y) denotes the coordinates of pixels in the natural shares and the secret image, $1 \le x \le \mathrm{w}$, $1 \le y \le \mathrm{h}$.
- (x_1, y_1) represents the coordinates of the left-top pixel in each block.
- $p_{\varphi}^{x,y}$ denotes the value of color φ, $\varphi \in \{\mathrm{R}, \mathrm{G}, \mathrm{B}\}$ for pixel (x, y) in natural share N, $0 \le p_{\varphi}^{x,y} \le 255$.
- Pixel value $\mathrm{H}^{x,y}$ is the sum of RGB color values of pixel (x, y) in natural share N and

$$\mathrm{H}^{x,y} = p_R^{x,y} + p_G^{x,y} + p_B^{x,y}. \tag{1}$$

- M represents the median of all pixel values $(\mathrm{H}^{x_1,y_1}, \ldots, \mathrm{H}^{x_b,y_b})$ in a block of N.
- $\boldsymbol{F}$ is the feature matrix of N, the element $f^{x,y} \in \boldsymbol{F}$ denotes the feature value of pixel (x, y). If the feature value $f^{x,y}$ is 0, the feature of pixel (x, y) in N is defined as black. If $f^{x,y}$ is 1 the feature of pixel (x, y) in N is defined as white.

As Fig. 3 shows, the feature extraction module consists of three processes—binarization, stabilization, and chaos processes. First, a binary feature matrix is extracted from natural image N via the binarization process. Then, the stabilization balances the occurrence frequency of values 1 and 0 in the matrix. Finally, the chaos process scatters the clustered feature values in the matrix.

Fig. 3. The block diagram of the feature extraction.

In the binarization process, the binary feature value of a pixel can be determined by a simple threshold function $\mathcal{F}$ with a set threshold. To obtain an approximate appearance probability for binary values 0 and 1, the median value M of pixels in the same block is an obvious selection as the threshold. Hence, for each block, the extraction function of pixel (x, y) of N is defined as follows:

$$f^{x,y} = \mathcal{F}(\mathrm{H}^{x,y}) = \begin{cases} 1, & \mathrm{H}^{x,y} \ge M, \\ 0, & \text{otherwise}. \end{cases} \tag{2}$$

The stabilization process is used to balance the number of black and white pixels of an extracted feature image in each block. The number of unbalanced black pixels $\mathrm{Q_s}$ can be calculated as follows:

$$\mathrm{Q_s} = \left(\sum_{\substack{\forall x_1 \le x \le x_b \\ \forall y_1 \le y \le y_b}} f^{x,y} \right) - \frac{b^2}{2}. \tag{3}$$

In the process, there are $\mathrm{Q_s}$ pixels in which $f^{x,y} = 1$ is randomly selected and then the value of these pixels is set to 0. The process ensures that the number of black and white pixels in each block is equal.

In a natural image, pixels with the same or approximately the same values may cluster together in a continuous region. These clustered pixels have the same feature value; hence it will lead to the feature image and to the generated share revealing some textures of the natural image in the subsequent encryption process. The chaos process is used to eliminate the texture that may appear on the extracted feature images and the generated share. In the process, the original feature matrix will be disordered by adding noise in the matrix. First, randomly select $\mathrm{Q_c}$ black feature pixels ($f^{x,y} = 0$) and $\mathrm{Q_c}$ white feature pixels ($f^{x,y} = 1$) from each block, then alter the value of these pixels. That is, the feature value of a pixel at coordinates (x, y) will be changed to 0 when $f^{x,y} = 1$, and vice versa. Assume $\mathrm{P_{noise}}$ be the probability to add noise in the matrix, the value of $\mathrm{Q_c}$ is as follows:

$$\mathrm{Q_c} = \mathrm{b}^2/2 \times \mathrm{P_{noise}}. \tag{4}$$

The feature extraction algorithm (FE) is shown in Algorithm 1. The algorithm extracts one binary feature matrix F from each input natural image N. Steps 3–5 determine the feature values of each pixel in a block. The stabilization process is performed in Steps 6 and 7. Steps 8–11 add noise based on a given parameter $\mathrm{P_{noise}}$. Step 12 outputs feature matrix $\boldsymbol{F}$.

Example 1. Suppose pixel value $\mathrm{H}^{x,y}$, $1 \le x, y \le 4$, in a block with 4×4 pixels, as shown in Fig. 4(a), and $\mathrm{P_{noise}} = 0.5$. Median value M of the block is 185. There are 6 pixel values less than M, hence the feature values at coordinates (4, 2), (2, 3), (3, 2), (1, 4), (2, 4), and (3, 4) are 0. The feature values at other coordinates are set to 1. By Eq. (3), we have

Algorithm 1 Feature Extraction Algorithm (FE)

Algorithm FE()
Input: N, b, P_{noise}
Output: ***F***

1. Divide N into blocks with $b \times b$ pixels
2. For each block repeat Steps 3—11
3. $\forall x_1 \le x \le x_b, y_1 \le y \le y_b$, calculate $H^{x,y}$ by Eq. (1)
4. Calculate M
5. $\forall x_1 \le x \le x_b, y_1 \le y \le y_b$, determine $f^{x,y}$ by Eq. (2)
6. Calculate Q_s by Eq. (3)
7. Randomly select Q_s pixels where $f^{x,y} = 1$ and $H^{x,y} = M$, let $f^{x,y} \leftarrow 0$
8. Calculate Q_c by Eq. (4)
9. Randomly select Q_c candidate pixels where $f^{x,y} = 1$
10. Randomly select Q_c candidate pixels where $f^{x,y} = 0$
11. Alter all values of $f^{x,y}$ that were selected in Steps 9 and 10
12. Output ***F***

250	200	200	200
185	200	200	160
185	60	90	185
110	80	120	190

(a)

1	1	1	1
0	1	1	0
1	0	0	0
0	0	0	1

(b)

<u>0</u>	1	1	<u>0</u>
0	<u>0</u>	<u>0</u>	<u>1</u>
1	<u>1</u>	<u>1</u>	0
<u>1</u>	0	0	1

(c)

Fig. 4. An example of the feature extraction process: (a) pixel values in a 4 × 4 block, (b) the resultant feature matrix in the stabilization process, (c) the resultant matrix in the feature extraction process.

$Q_s = 10 - 8 = 2$, and two feature values at coordinates $(1, 2)$, $(1, 3)$, and $(4, 3)$ must be set to 0. The resultant matrix in Step 7 is shown in Fig. 4(b). In this matrix, the feature values 0 and 1 are balanced. However, the extracted features remain clustered together; for example, coordinates $(1, 1)$, $(2, 1)$, $(3, 1)$, $(4, 1)$, $(2, 2)$, and $(3, 2)$ have the same feature value. Finally, by Eq. (4), there are $Q_c = 4^2/2 \times 0.5 = 4$ coordinates with feature value 1 (0) that must be altered to 0 (1). The resultant matrix in Steps 9–11 is shown in Fig. 4(c), with the altered feature values underlined. In this matrix, the clustering has been reduced.

The feature matrix has the following properties:

Property 1. The values 0 and 1 in the extracted binary feature matrix have the same appearance probability (i.e., 0.5 for each).

Proof. This property can be achieved by the stabilization process (i.e., Steps 6 and 7 in Algorithm FE). □

Property 2. The feature matrix depends on the contents of the corresponding natural image rather than the secret images.

Property 2 indicates an important feature; that is, no one can decrypt the secret from the natural shares.

2) The Image Preparation and Pixel Swapping Processes

The image preparation and pixelswapping processes are used for preprocessing printed images and for post-processing the feature matrices that are extracted from the printed images. The printed images were selected for sharing secret images, but the contents of the printed images must be acquired by computational devices and then be transformed into digital data.

The suggested flow of the image preparation process is shown in Fig. 5. In the first step, the contents of the printed

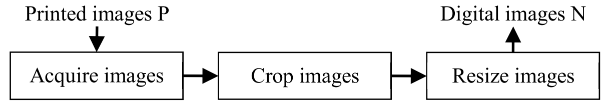


Fig. 5. Flow of the image preparation process.

(a) (b)

Fig. 6. An example of the image preparation process: (a) a hand-painted picture (3264 × 2448 pixels) was captured by the digital camera on the iPhone 4S, (b) the resultant picture (512 × 512 pixels).

images can be acquired by popular electronic devices, such as digital scanners and digital cameras. To reduce the difference in the content of the acquired images between the encryption and decryption processes, the type of the acquisition devices and the parameter settings (e.g., resolution, image size) of the devices should be the same or similar in both processes. The next step is to crop the extra images. Finally, the images are resized so they have the same dimensions as the natural shares. An example of the image preparation process is illustrated in Fig. 6. The hand-painted picture is drawn on A4 paper. First, the picture is captured using a popular smart phone, Apple iPhone 4S, as shown in Fig. 6(a). The picture then is processed using the Paint application in Microsoft Windows 7. Eventually, the picture is cropped and resized as a rectangular image as shown in Fig. 6(b). The resultant picture is used in the experiments in the subsequent section.

Because of the distortions introduced into the acquired digital images during the image preparation process, different distortions are caused by each the encryption process and the decryption process. In other words, the acquired digital images in the encryption and decryption phases are not the same. These distortions result in noise that appears in the recovered images. When a large amount of noise clusters together, the image is severely disrupted, which may makes it impossible for the naked eye to identify it. The pixels-wapping process is used to cope with this problem.

After the feature extraction process, a pixel-swapping module is applied to randomize the original spatial correlation of pixels in a printed image. The module pseudo-randomly exchanges the feature values of a pair of coordinates in a feature matrix. The permuted pixel sequence is determined by the random number generator. After the process, the distortions that were introduced in the image preparation process were spread in a feature matrix, and the noise also is distributed in the recovered image rather than clustered together. If the noise is distributed uniformly, the human visual system has a higher probability of recognizing the recovered image. In other words, the pixels-wapping module promotes tolerance of the image distortion caused by the image preparation process.

B. Encryption/Decryption Algorithms

The proposed (n, n)-NVSS scheme can encipher a true-color secret image by $n - 1$ innocuous natural shares and one noise-like share. For one image, we denote a bit with the same weighted value in the same color as a bit plane; then a true color secret image has 24 bit-planes. Thus, the feature images and the noise-like share also are extended to 24 bit-planes. Each bit-plane of a feature image consists of a binary feature matrix that corresponds to the same bit-plane as the secret image.

Before encryption (resp. decrypt) of each bit-plane of the secret image, the proposed algorithm first extracts $n-1$ feature matrices from $n-1$ natural shares. Then the bit-plane of the secret image (resp. noise-like share) and $n-1$ feature matrices execute the XOR operation (denoted by $\oplus$) to obtain the bit-plane of the share image (resp. recovered image). Therefore, to encrypt (resp. decrypt) a true-color secret image, the encryption (resp. decryption) procedure must be performed iteratively on the 24 bit-planes.

The notations used in the NVSS encryption/decryption process are defined as follows:

- φ denotes a color plane of an image, $\varphi \in \{\mathrm{R, G, B}\}$.
- S is the input image; S_φ denotes an element of S in color-plane φ.
- $\overline{\mathrm{S}}$ is the output image; $\overline{\mathrm{S}}_\varphi$ denotes an element of $\overline{\mathrm{S}}$ in color-plane φ.
- FI_α denotes a feature image of natural share N_α.
- $\mathrm{FI}_{\alpha,\varphi}$ denotes an element of feature images in color-plane φ.
- $p_{\alpha,\varphi}^{x,y}$ denotes the pixel value of $\mathrm{FI}_{\alpha,\varphi}$ at coordinates (x, y), $0 \le p_{\alpha,\varphi}^{x,y} \le 255$.
- ρ is the seed of the random number generator G.
- t is the amount of pixel swapping for a feature image of a printed image.

Algorithm 2 (n, n)-NVSS Encryption/Decryption Algorithm

Algorithm NVSS()
Input: S, $\mathrm{N}_1, \ldots, \mathrm{N}_{\mathrm{n_p+n_d}}$, $\mathrm{n_p}$, $\mathrm{n_d}$, b, $\mathrm{P_{noise}}$, ρ, t
Output: $\overline{\mathrm{S}}$

1. Initialize the random number generator G by the seed ρ
2. $n \leftarrow \mathrm{n_p} + \mathrm{n_d} + 1$
3. $\forall 1 \le \alpha < n, \forall \varphi \in \{R, G, B\}, \mathrm{FI}_{\alpha,\varphi} \leftarrow 0$
4. $\forall 1 \le \alpha < n, \forall \varphi \in \{R, G, B\}, \forall 0 \le i \le 7$, **repeat** Steps 5 and 6
5. **Call** procedure FE(N_α, b, $\mathrm{P_{noise}}$, $\boldsymbol{F}$)
6. $\forall (x, y), x \in [1, w], y \in [1, h], p_{\alpha,\varphi}^{x,y} \leftarrow p_{\alpha,\varphi}^{x,y} + f^{x,y} \times 2^i$
7. **If** $\mathrm{n_p} = 0$ **then goto** Step 12
8. $\forall 1 \le \alpha \le \mathrm{n_p}$, **repeat** Steps 9—11 t times
9. Randomly selects (x_1, y_1), $x_1 \in [1, w]$, $y_1 \in [1, h]$
10. Randomly selects (x_2, y_2), $x_2 \in [1, w]$, $y_2 \in [1, h]$
11. $\forall \varphi \in \{R, G, B\}$, exchange values of $p_{\alpha,\varphi}^{x_1,y_1}$ and $p_{\alpha,\varphi}^{x_2,y_2}$
12. $\forall \varphi \in \{R, G, B\}, \overline{\mathrm{S}}_\varphi \leftarrow \mathrm{S}_\varphi \oplus \mathrm{FI}_{1,\varphi} \oplus \cdots \oplus \mathrm{FI}_{n-1,\varphi}$
13. **Output** $\overline{\mathrm{S}}$

Algorithm 2 lists the encryption/decryption algorithms. The input natural shares $(\mathrm{N}_1, \ldots, \mathrm{N}_{n-1})$ of the scheme include $\mathrm{n_p}$ printed images and $\mathrm{n_d}$ digital images ($\mathrm{n_p} \ge 0$, $\mathrm{n_d} \ge 0$, $\mathrm{n_p} + \mathrm{n_d} \ge 1$, and $n = \mathrm{n_p} + \mathrm{n_d} + 1$). The $\mathrm{n_p}$ printed images must be processed and transformed into digital form in the image preparation process. All input images are 24-bit/pixel true-color images. Step 1 initializes random number generator G by seed ρ. Function G is used for the feature extraction and pixels-wapping processes. The encryption and decryption processes use the same seed ρ to generate an identical pseudo-random sequence. Step 3 initializes all feature images; that is, resets all pixel values $p_{\alpha,\varphi}^{x,y}$ in all feature images FI_α. Steps 4–6 extract one feature image with a 24-bit depth per pixel from each natural share. Step 5 extracts a binary feature matrix from a natural share by calling algorithm FE. Step 6 adds the extracted matrix to corresponding bit and color planes of a feature image. Steps 8–11 perform the pixel-swapping process for each feature image extracted from the printed images. For each feature image, the pixel-swapping process randomly selects a pair of pixels in a feature image in Steps 9 and 10, and then swaps the values of two pixels in Step 11. Step 12 stacks input image S and feature images $\mathrm{FI}_1, \ldots, \mathrm{FI}_{n-1}$ by applying the XOR operation in each color plane. Finally, the resultant image $\overline{\mathrm{S}}$ is the output in Step 13. The pseudo code of the algorithm is for true-color secret images; however it is also applicable for 8-bit gray and binary images. In the case of an encryption/decryption gray (binary) secret image, the algorithm extracts an 8-bit (1-bit) feature image for each natural image to fit the information quantity of the secret image.

There are two main loops in the algorithm: Steps 4–6 and Steps 8–11. The time complexity of algorithm FE is O(hw). The time complexity of Step 4–6 is $\mathrm{O}(n\mathrm{c_d}\mathrm{hw})$, where $\mathrm{c_d}$ represents the color depth of a pixel. The value of $\mathrm{c_d}$ depends on the color depth of the secret image. For example, if the secret image is with 24-bit/pixel color depth, the value of $\mathrm{c_d}$ is 24. The value of $\mathrm{c_d}$ is 8, while the secret image is a gray image with 8-bit gray levels. The time complexity of Steps 8–11 is $\mathrm{O}(\mathrm{n_p}\mathrm{thw})$. Hence, the complexity of the encryption/decryption algorithm is $max\left\{\mathrm{O}\left(n\mathrm{c_d}\mathrm{hw}\right), \mathrm{O}(\mathrm{n_p}\mathrm{thw})\right\}$.

The algorithm in Algorithm II can be used for the encryption and decryption phases by setting various parameters as follows:

- Encryption: Input images include $n-1$ natural shares and one secret image. The output image is a noise-like share.
- Decryption: Input images include $n-1$ natural shares and one noise-like share. The output image is a recovered image.

The proposed encryption algorithm has the properties discussed below.

Property 3. The amount of information required for the generated share is the same as for the secret image.

Proof. In the encryption process of the algorithm, one binary feature value must be extracted from one natural share to share 1 bit of a secret pixel. Each pixel in the generated share is yielded by XOR-ing the corresponding secret pixel and $n-1$ binary feature values that were extracted from $n-1$ natural shares. Therefore, the generated share has the same amount of information as the secret image. □

Property 3 shows that the generated share in the proposed (n, n)-NVSS scheme is free of pixel expansion.

Property 4. Pixel values in a feature image are distributed uniformly over $[0, 255]$.

Proof. As proved in Property 1, values 0 and 1 share the same appearance probability in a feature bit. In Step 6 of the NVSS encryption/decryption algorithm, an 8-bit feature value $p_{\alpha,\varphi}^{x,y}$ is composed of 8 various feature bits that were extracted from N_α in 8 iterations. Each bit plane in $p_{\alpha,\varphi}^{x,y}$ also contains 50% value 0 and 50% value 1. This property is the same as the binary number system. Hence, the value of $p_{\alpha,\varphi}^{x,y}$ will range between 0 and 255 and each value in the range will have the same appearance probability. The stacking operation uses a logical XOR operator; hence, the pixel values in the stacked feature image FI, $\mathrm{FI} = \mathrm{FI}_1 \oplus \cdots \oplus \mathrm{FI}_{n-1}$, remain in the same distribution. □

Property 5. Pixel values in a feature image are distributed randomly.

Proof. The binary feature values of a natural image are a function of the image content and a random number generator G. Pixel values in the natural image can be treated as a random sequence with $h \times w$ samples, but the image contents are unpredictable. Hence, the binary features are random. Pixel values in the feature image are composed of 8 binary feature matrices, so the pixel values are distributed randomly. □

Property 6. The generated share is secure.

Proof. The share was generated by stacking a secret image and n-1 feature images. As proved in Property 4 and Property 5, pixel values in each feature image are distributed randomly and uniformly. These feature images can be treated as $n-1$ one-time pads. The length of each one-time pad is equal to the length of the secret image. The encryption operation uses the logical XOR operator. In cryptography, an encryption process with the above-mentioned features has been proven impossible to crack. Hence, the generated share is secure. □

C. Hide the Noise-Like Share

In this section, steganography and the Quick-Response Code (QR code) techniques are introduced to conceal the noise-like share and further reduce intercepted risk for the share during the transmission phase.

In the proposed NVSS scheme, a dealer can hide the generated share by using existing steganography. The amount of information that can be hidden in a cover image is limited and depends on the hiding method. To embed the generated share in a cover image, generally the dimension of the cover image must be larger than that of the secret image. If the share can be hidden in the cover image and then can be retrieved totally, the secret image can be recovered without distortion. We leave the details of using steganography to hide shares to the reader; our focus is on how to hide the share in printed media using QR code technology.

The QR code is a two-dimensional code first designed for the automotive industry by DENSO WAVE in 1994 [19]. The QR code, which encodes meaningful information in both dimensions and in the vertical and horizontal directions, can carry up to several hundred times the amount of data carried by barcodes. The code is printed on physical material and can be read and decoded by various devices, such as barcode readers and smart phones. Today, the QR code is widely used in daily life, and is widely visible, on the surface of products, in commercial catalogs and flyers, in electronic media, and elsewhere. It is this ubiquitous nature of the QR code that makes it suitable for use as a carrier of secret communications.

0	1	1	0	1	0	1	0	The 1st decimal value
0	0	0	1	0	1	1	1	
1	1	1	0	1	1	0	1	The 2nd decimal value
1	0	0	1	0	0	0	0	

(a) (b)

Fig. 7. An example of the feature matrix to the QR code encoding: (a) the feature matrix, (b) the corresponding QR code of the matrix.

The amount of data that can be stored in the QR code symbol depends on the data type (e.g., numeric, alphanumeric, byte/binary, Kanji), version, and error correction level. In this paper, we will pretend the noise-like share as the numeric type of the QR code. The encoding process consists of two steps: First, transform pixels on the share into binary values and represent the values in a decimal format. In this step, 16 binary feature bits are converted into a 5-digit decimal value, which ranges from 0 to 65,535. Second, we encode the decimal values into QR code format. For example, the binary feature values of the upper matrix in Fig. 7(a), (MSB)1110100001010110_2 (LSB, the upper left corner of the upper matrix), can be represented as $59{,}478_{10}$. The lower matrix can be represented as $2{,}487_{10}$. Hence, a QR code corresponding to the numeric string 5947802487 can be generated as shown in Fig. 7(b).

Applying the above-mentioned method, 5 numeric characters can be used to encode 16 feature bits. Hence, the maximum capacity of the QR code (i.e., version 40, the error correction level: L) for encoding the noise-like share is $\lfloor 7,089/5 \rfloor \times 16 = 22,627$ bits. This limited capacity of the QR code means that a single QR code cannot encode a share which contains more than 22,627 data bits. A possible solution is to use multiple QR codes for encoding a share; however, placing too many QR codes together may arouse suspicion and incur risk during the transmission phase. We propose an alternative solution to address this problem.

Suppose C_H denotes the maximum capacity of the selected hiding media and C_s denotes the amount of information in the share, where $C_s \geq C_H$. The capacity ratio C_r is defined as $C_r = \lceil C_s/C_H \rceil$. $C_r = 1$ means the share can be totally hidden in the hiding media. If the share cannot be hidden in the media (i.e., $C_r > 1$), then C_r consecutive bits must be encoded into a single bit (called a stego-bit). The value of the stego-bit is decided by majority of the C_r bits.

The share-hiding algorithm is shown in Algorithm 3. The algorithm hides feature matrix $\boldsymbol{F}$ in a numeric string S_{QR}. Then the string can be encoded to the QR code (a stego-share) by QR code generators. Steps 1 and 2 initialize the parameters. Steps 3–6 reduce the amount of information in the feature matrix F to fit within the capacity of the hiding media C_H. These steps are performed only in case of $C_r > 1$. Step 3 removes C_r consecutive bits from the front of string S_F by calling procedure *remove*(). Step 4 decides the value of stego-bit s_b by majority. Function H(S) represents the Hamming weight of bit string S. Then, the stego-bit is appended to bit string F_{QR} in Step 5. Steps 7–12 convert F_{QR} to the numeric

Algorithm 3 The Share-Hiding Algorithm

```
Algorithm EM_SQR()
Input: F
Output: S_QR
1.  C_s ← h × w; C_r ← ⌈C_s/C_H⌉; F_QR ←null string
2.  Vectorize F to bit-string S_F
3.  S_t ←remove(S_F, C_r)
4.  If H(S_t) < C_r/2 then s_b ← 0 else s_b ← 1
5.  Append s_b to F_QR
6.  If S_F ≠null string then goto step 3
7.  S_QR ←null string
8.  Convert h, w, C_r to strings and append these strings to S_QR
9.  S_t ←remove(F_QR, 16)
10. num ← Σ_{0≤i≤15} S_{t,i} × 2^i
11. Convert num to string and append the string to S_QR
12. If F_QR ≠null string then goto step 9
13. Output S_QR
```

string S_{QR}. Notation $S_{t,i}$ represents the binary value of the i-th character of string S_t. Finally, output S_{QR} in Step 13.

Algorithm 4 The Share Extraction Algorithm

```
Algorithm EX_SQR()
Input: S_QR
Output: F
1. Retrieve h, w, and C_r from S_QR
2. F_QR ←null string
3. S ←remove(S_QR, 5)
4. num ←str2int(S)
5. ∀0 ≤ i ≤ 15, append C_r consecutive bits with value num_i to F_QR
6. If S_QR ≠null string then goto step 3
7. Transform vector F_QR to matrix F with h × w entries
8. Output F
```

In the decryption phase, on receipt of the stego-share, the hidden information must be extracted from the stego-share. Algorithm 4 lists the share extraction algorithm that extracts a feature matrix F from numeric string S_{QR}. The string S_{QR} is decoded from the stego-share, which is in QR code format. Step 1 retrieves the related parameters from S_{QR}. Steps 3–5 transform 5 numeric characters into binary form. Step 3 removes 5 consecutive numeric characters from the front of number string S_{QR} by calling procedure *remove*(). Step 4 converts string S to its integer value by procedure *str2int*(). Step 5 transforms the value to a corresponding binary bit string and appends it to bit string F_{QR}. Notation num_i represents the i-th bit of integer num. Step 7 converts F_{QR} to the resultant feature matrix F. Step 8 outputs feature matrix $\boldsymbol{F}$.

The share-hiding algorithm alters some of the pixel values so that they fit within the capacity of the stego-share in the encryption phase. These altered pixels will introduce noise into the recovered image in the decryption phase. The following property analyzes the relationship between the capacity ratio, C_r, and the contrast value of the recovered binary image.

Property 7. Suppose P_1 (P_0) represents the appearance probability of altering a black (white) pixel to white (black) in the reconstructed share. The contrast of the recovered binary image, α, can be defined as $\alpha = 1 - 2(P_0 + P_1)$, where $P_1 = \sum_{i=b_h}^{C_r-1}\left(\frac{C_r-i}{C_r} \times \frac{\binom{C_r}{i}}{2^{C_r}}\right)$ and $P_0 = \sum_{i=1}^{b_\ell}\left(\frac{i}{C_r} \times \frac{\binom{C_r}{i}}{2^{C_r}}\right)$, $b_h = \lceil C_r/2 \rceil$, $b_\ell = \lceil C_r/2 - 1 \rceil$, and $C_r > 1$.

Proof: Based on Step 4 in the share-hiding algorithm, stego-bit s_b of bit string S_t can be determined by the following function

$$s_b = \begin{cases} 0, & H(S_t) < C_r/2 \\ 1, & \text{otherwise} \end{cases}.$$

If $i = H(S_t)$ and $i < C_r/2$, then we have $s_b = 0$ and i pixels will be altered in C_r reconstructed bits. The probability that the altered pixel will appear in a white reconstructed pixel is $P_0 = \sum_{i=1}^{b_\ell}\left(\frac{i}{C_r} \times \frac{\binom{C_r}{i}}{2^{C_r}}\right)$, where $b_\ell = \lceil C_r/2 - 1 \rceil$. On the contrary, if $i \geq C_r/2$, then we have $s_b = 1$ and $(C_r - i)$ pixels will be altered in C_r reconstructed pixels. The probability that the altered pixel will appear in a black reconstructed pixel is $P_1 = \sum_{i=b_h}^{C_r-1}\left(\frac{C_r-i}{C_r} \times \frac{\binom{C_r}{i}}{2^{C_r}}\right)$, where $b_h = \lceil C_r/2 \rceil$. The overall probability that the altered pixel will appear in a reconstructed bit is $P_0 + P_1$. In the proposed NVSS scheme, the altered pixels in a reconstructed share will decrease the blackness level of the recovered black pixels from 1 to $1-(P_0 + P_1)$ and increase the blackness level of the recovered white pixels from 0 to $(P_0 + P_1)$. Hence, the contrast of the recovered image is $\alpha = 1 - 2(P_0 + P_1)$. □

The proposed share-hiding algorithm can be applied to hide the noise-like share in the QR code and digital media. Regardless the types of cover media, the contrast of recovered images will obey Property 7. Hence, the major concern about the contrast of recovered images is the capacity of the media for hiding a noise-like share. The capacity of the QR code depends on the selected QR code version. To hide the share in digital media, we should select a specific steganography (e.g., the perturbed quantization (PQ) steganography [20], etc.) first, and apply it to hide information in the media. Then, use the proposed share-hiding algorithm to hide the noise-like share. Hence, the capacity for hiding information depends on the usage of steganography technique and the digital media.

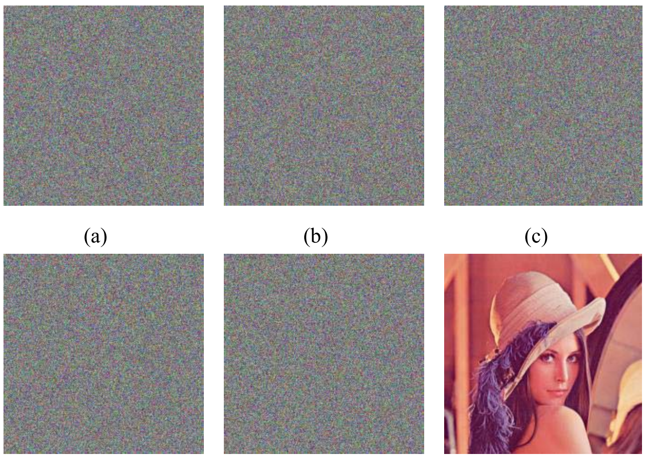
For example, if we would like to hide a binary share with $h \times w$ pixels in a 24-bits color image, which is in the JPEG format with $2h \times 2w$ pixels. The PQ steganography, which is a high security method for embedding information into the JPEG images, is used for hiding the share. Suppose the compression ratio of the cover image is 10:1 and the embedding rate of the PQ steganography is 0.2, there are $0.1 \times 0.2 \times 24 \times 2h \times 2w = 1.92 \times h \times w$ bits in the cover image for hiding the binary share; therefore, capacity ratio C_r is smaller than 1 and the secret image can be reconstructed perfectly. If the share is an 8-bits gray image, the capacity ratio C_r approaches to 4.2. Based on Property 7, the contrast of the recovered image is around 37.5%. While embedding rate is enlarged to 0.3, the contrast of the recovered image will be promoted to 50%.

V. Experiments

In this section, we perform three experiments to evaluate the performance of the proposed NVSS scheme.

(a) (b) (c)

Fig. 8. The natural shares (N_1, N_2, and N_3) in a $(4, 4)$-NVSS scheme: (a) N_1, (b) N_2, (c) N_3.



(a) (b) (c)

(d) (e) (f)

Fig. 9. Experimental results of Experiment-I: (a) share S, (b) $S \oplus FI_1$, (c) $S \oplus FI_1 \oplus FI_2$, (d) $S \oplus FI_1 \oplus FI_3$, (e) $S \oplus FI_2 \oplus FI_3$, (f) recovered image of SE_1 ($S \oplus FI_1 \oplus FI_2 \oplus FI_3$).

A. Experiment I

This subsection demonstrates the performance of the proposed NVSS scheme in the case of a (4, 4)-NVSS scheme. Fig. 8 shows three natural shares in the experiments. The secret image SE_1 is the well-known picture—"Lena" (as shown in Fig. 9(f)). All natural shares are taken from travel photos of tourists. These images are in true color format and their dimensions are 512×512 pixels. Parameters b and P_{noise} are set to 8 and 0.5, respectively.

Fig. 9(a) shows share S yielded by the (4, 4)-NVSS scheme. The appearance of S consists of a large amount of random pixels and the textures are concealed, which indicates that Algorithm FE is very efficient at eliminating textures in the share Fig. 9(b) to Fig. 9(f) provide examples of reconstructed images obtained by stacking noise-like share S and the feature images (denoted as FI). Fig. 9(f) is the perfectly recovered image decrypted by stacking share S and all feature images (i.e., FI_1, FI_2, and FI_3). The other images (i.e., Fig. 9(b) to Fig. 9(e)) cannot reveal any texture related to the secret image or the natural shares because one or more of the natural shares is missing.

Fig. 10 is the graphic representation showing the statistical results on the distribution of pixel values in share S and secret image (SE_1, Lena). The distributions in SE_1 in the red, green, and blue color planes are denoted as Secret (R), Secret (G), and Secret (B), respectively. Fig. 10 shows that the distribution in S (denoted as Share in Fig. 10, in each color plane is random;

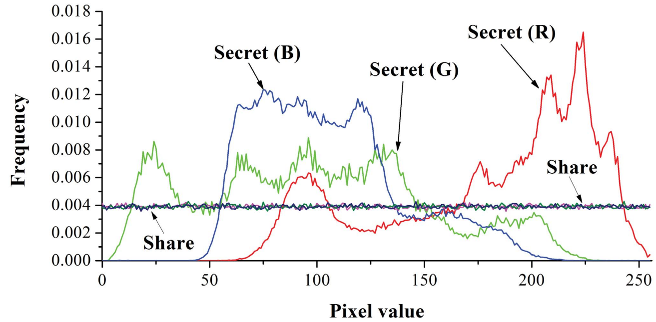


Fig. 10. The distribution of pixel value in share S and secret image SE_1.

it is totally different from the distribution in SE_1. Hence, it is difficult to obtain any information related to SE_1 from share S. The distribution in share S also agrees with Property 5.

B. Experiment II

This experiment evaluates the performance of the proposed NVSS scheme for sharing color and binary secret images by diverse shares. The shares used in the (5, 5)-NVSS scheme include three digital images and one hand-painted picture. The digital images and the color secret image are the same as those used in Experiment I. The hand-painted picture is drawn on A4 paper, as shown in Fig. 6(a). The picture is processed by the image preparation process to obtain two digitized shares, one for the encryption process and one for the decryption process. The digitized share for the encryption process is captured by a digital camera—Canon 650D, as shown in Fig. 10(a). Another digitized share is captured by the digital camera on a smart phone, iPhone4S, as shown in Fig. 10(b). After the image preparation process, all images in the experiments are 512×512 pixels. Parameters b and P_{noise} for all digital shares are set to 8 and 0.5, respectively. Parameter b for the printed share is set to 256.

Fig. 11(c) shows the difference between Fig. 11(a) and Fig. 11(b). The difference can be evaluated by the quantitative metric—PSNR (the peak signal to noise ratio). The PSNR value between Fig. 11(a) and Fig. 11(b) is only 19.74 dB. The low PSNR means that high distortions are introduced into the digitized share that will be used in the decryption phase. The distortion is introduced by various devices in capturing the hand-painted share during the encryption/decryption phases.

Fig. 11(d) and Fig. 11(f) illustrate the recovered image of the NVSS scheme for sharing color image SE_1 and binary image SE_2 (as shown in Fig. 11(e)). Although applying a printed share for the image sharing, noise will be introduced to the recovered image; the display quality of the recovered image remains clear enough to be recognized by the naked eye. The result is that the proposed NVSS scheme can use printed images as sharing images. This outcome is significant and has not been presented in previous research.

C. Experiment III

This subsection demonstrates the performance of the (4, 4)-NVSS scheme by using the QR code to hide the noise share. The natural shares are as shown in Fig. 8(a) to Fig. 8(c).

Fig. 11. Experimental results of Experiment-II: (a) the digitalized share that is captured by Canon 650D digital camera, (b) the digitalized share that is captured by iPhone 4S, (c) difference between Fig. 11(a) and Fig. 11(b), (d) recovered image of SE_1 (PSNR = 12.12 dB), (e) binary secret image (SE_2), (f) recovered image of SE_2 (contrast = 60.64%).

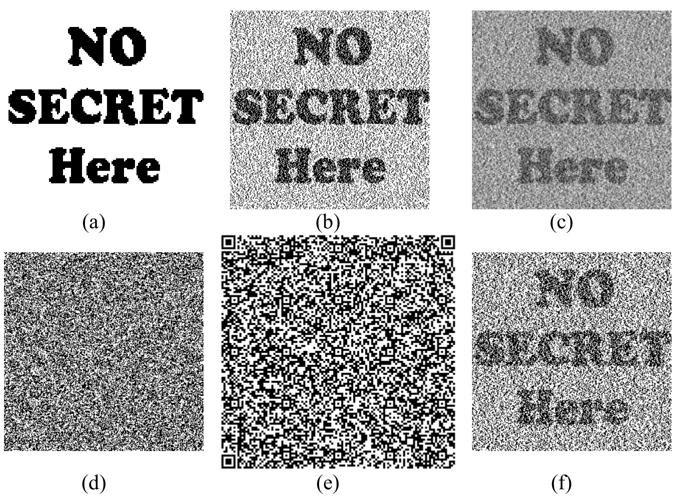


Fig. 12. Experimental results of Experiment III: (a) recovered image (128 × 128 pixels, 96 dpi, contrast = 100%), (b) recovered image (256 × 256 pixels, 192 dpi, contrast = 50.11%), (c) recovered image (512 × 512 pixels, 384 dpi, contrast = 22.65%), (d) noise share (256 × 256 pixels, 192 dpi), (e) the corresponding QR code (version 25, the error correction level: L) of Fig. 12 (d), (f) recovered image (256 × 256 pixels, 192 dpi, contrast = 31.32%), which adopts the QR code in Fig. 12(e) as the stego-share.

The secret images are binary and have the same cipher text—"NO SECRET Here". Parameters b and P_{noise} are set to 8 and 0.5, respectively.

First, we investigate the display quality of the recovered images by using the QR code with the maximum capacity (i.e., version 40) to hide the noise share. Fig. 12(a), (b), and (c) are the recovered images for three images with various dimensions: 128×128, 256×256, and 512×512 pixels. Fig. 12(a) shows that a lossless recovered image can be obtained when the QR code can hide the noise share entirely. When the amount of information of the share increases, the contrast of the recovered images remains at an acceptable level. As shown in Fig. 12(c), the recovered image can be recognized by the naked eye easily even if the dimension of the image is as large as 512×512 pixels.

TABLE I
THE CONTRAST OF RECOVERED IMAGE BY USING VARIOUS VERSIONS OF QR CODES (IMAGE SIZE: 256 × 256 PIXELS, THE ERROR CORRECTION LEVEL OF THE QR CODE IS L)

C_r	Contrast (%)		The preferred versions
	Theoretical	Experimental	
3	50.00	50.11	40
4, 5	37.50	37.21	30—39
6, 7	31.25	31.32	25—29
8, 9	27.34	27.69	22—24
10, 11	24.61	25.00	19—21
12	22.56	22.50	18
14, 15	20.95	20.90	16—17

Second, we explore the capability of various versions of the QR code to hide the noise share. The secret image used in this experiment is a binary image of size 256×256 pixels. Fig. 12 (d) to Fig. 12(f) demonstrate the implementation results of a noise share and its corresponding QR code (version 25) and the recovered image in the $(4, 4)$-NVSS scheme. Table I lists all available QR code versions that can be used to reconstruct secret images with a contrast value greater than 20%. The results indicate that the experimental contrast values are very close to the theoretical values calculated by Property 7. The high contrast of the recovered images and the wide range of QR code versions demonstrate the capability of the QR code.

VI. CONCLUSION

The paper proposes a VSS scheme, (n, n)-NVSS scheme, that can share a digital image using diverse image media. The media that include $n-1$ randomly chosen images are unaltered in the encryption phase. Therefore, they are totally innocuous. Regardless of the number of participants n increases, the NVSS scheme uses only one noise share for sharing the secret image. Compared with existing VSS schemes, the proposed NVSS scheme can effectively reduce transmission risk and provide the highest level of user friendliness, both for shares and for participants.

This study provides four major contributions. First, this is the first attempt to share images via heterogeneous carriers in a VSS scheme. Second, we successfully introduce hand-printed images for images-haring schemes. Third, this study proposes a useful concept and method for using unaltered images as shares in a VSS scheme. Fourth, we develop a method to store the noise share as the QR code.

ACKNOWLEDGMENT

Hereby, the authors appreciate the anonymous reviewers for their valuable comments.

REFERENCES

[1] M. Naor and A. Shamir, "Visual cryptography," in *Advances in Cryptology*, vol. 950. New York, NY, USA: Springer-Verlag, 1995, pp. 1–12.
[2] R. Z. Wang, Y. C. Lan, Y. K. Lee, S. Y. Huang, S. J. Shyu, and T. L. Chia, "Incrementing visual cryptography using random grids," *Opt. Commun.*, vol. 283, no. 21, pp. 4242–4249, Nov. 2010.

[3] P. L. Chiu and K. H. Lee, "A simulated annealing algorithm for general threshold visual cryptography schemes," *IEEE Trans. Inf. Forensics Security*, vol. 6, no. 3, pp. 992–1001, Sep. 2011.

[4] K. H. Lee and P. L. Chiu, "Image size invariant visual cryptography for general access structures subject to display quality constraints," *IEEE Trans. Image Process.*, vol. 22, no. 10, pp. 3830–3841, Oct. 2013.

[5] G. Ateniese, C. Blundo, A. D. Santis, and D. R. Stinson, "Extended capabilities for visual cryptography," *Theoretical Comput. Sci.*, vol. 250, nos. 1–2, pp. 143–161, Jan. 2001.

[6] C. N. Yang and T. S. Chen, "Extended visual secret sharing schemes: Improving the shadow image quality," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 21, no. 5, pp. 879–898, Aug. 2007.

[7] K. H. Lee and P. L. Chiu, "An extended visual cryptography algorithm for general access structures," *IEEE Trans. Inf. Forensics Security*, vol. 7, no. 1, pp. 219–229, Feb. 2012.

[8] Z. Zhou, G. R. Arce, and G. D. Crescenzo, "Halftone visual cryptography," *IEEE Trans. Image Process.*, vol. 15, no. 8, pp. 2441–2453, Aug. 2006.

[9] Z. Wang, G. R. Arce, and G. D. Crescenzo, "Halftone visual cryptography via error diffusion," *IEEE Trans. Inf. Forensics Security*, vol. 4, no. 3, pp. 383–396, Sep. 2009.

[10] I. Kang, G. R. Arce, and H. K. Lee, "Color extended visual cryptography using error diffusion," *IEEE Trans. Image Process.*, vol. 20, no. 1, pp. 132–145, Jan. 2011.

[11] F. Liu and C. Wu, "Embedded extended visual cryptography schemes," *IEEE Trans. Inf. Forensics Security*, vol. 6, no. 2, pp. 307–322, Jun. 2011.

[12] T. H. Chen and K. H. Tsao, "User-friendly random-grid-based visual secret sharing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 21, no. 11, pp. 1693–1703, Nov. 2011.

[13] T. H. N. Le, C. C. Lin, C. C. Chang, and H. B. Le, "A high quality and small shadow size visual secret sharing scheme based on hybrid strategy for grayscale images," *Digit. Signal Process.*, vol. 21, no. 6, pp. 734–745, Dec. 2011.

[14] D. S. Tsai, G. Horng, T. H. Chen, and Y. T. Huang, "A novel secret image sharing scheme for true-color images with size constraint," *Inf. Sci.*, vol. 179, no. 19, pp. 3247–3254, Sep. 2009.

[15] X. Wu, D. Ou, Q. Liang, and W. Sun, "A user-friendly secret image sharing scheme with reversible steganography based on cellular automata," *J. Syst. Softw.*, vol. 85, no. 8, pp. 1852–1863, Aug. 2012.

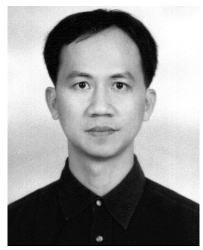
[16] C. Guo, C. C. Chang, and C. Qin, "A multi-threshold secret image sharing scheme based on MSP," *Pattern Recognit. Lett.*, vol. 33, no. 12, pp. 1594–1600, Sep. 2012.

[17] A. Nissar and A. H. Mir, "Classification of steganalysis techniques: A study," *Digit. Signal Process.*, vol. 20, no. 6, pp. 1758–1770, Dec. 2010.

[18] P. L. Chiu, K. H. Lee, K. W. Peng, and S. Y. Cheng, "A new color image sharing scheme with natural shadows," in *Proc. 10th WCICA*, Beijing, China, Jul. 2012, pp. 4568–4573.

[19] (2013). *QR Code.com* [Online]. Available: http://www.qrcode.com/en/index.html (Accessed).

[20] J. Fridrich, M. Goljan, and D. Soukal, "Perturbed quantization steganography with wet paper codes," in *Proc. Workshop Multimedia Sec.*, Magdeburg, Germany, Sep. 2004, pp. 4–15.



Kai-Hui Lee received the Ph.D. degree in electronic engineering from the National Taiwan University of Science and Technology, Taipei, Taiwan, in 2002. He is a Professor in the Department of Computer Science and Information Engineering, Ming Chuan University, Taipei.

His current research interests include visual cryptography, wireless networks, and network resource managements.

Pei-Ling Chiu received the Ph.D. degree in information management from the National Taiwan University, Taipei, Taiwan, in 2007. She is a Professor in the Department of Risk Management and Insurance, Ming Chuan University, Taipei.

Her research focuses on visual cryptography, wireless sensor networks, and optimizing technologies.